

DiffGenMed-VQA: A Diffusion Language Model–Based Medical Visual Question Answering System

TERM REPORT **CENG415 SENIOR DESIGN PROJECT AND SEMINAR 1** **(Academic)**

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ABSTRACT

Abstract

This project aims to go beyond existing classification-based approaches in the field of medical visual question answering (Med-VQA) by developing a diffusion-based generative artificial intelligence model. The proposed DiffGenMed-VQA model will process multimodal inputs consisting of medical images and natural language questions to generate contextually accurate, explainable, and naturally worded answers with explicit reasoning. The originality of the project lies in introducing a new paradigm that combines the strengths of diffusion modeling with large language models (LLMs). While existing Med-VQA approaches restrict answers to predefined label sets, DiffGenMed-VQA produces open-ended, semantically rich, and reasoned responses through conditional diffusion processes. This study will be one of the first examples of diffusion-based generative modeling in medical artificial intelligence and will significantly enhance explainability in clinical knowledge inference.

The model to be developed within the scope of the project will be based on a diffusion-based generative architecture designed according to the principle of bidirectional conditioning. A visual encoder (e.g., ViT, Swin Transformer) and a text encoder (e.g., BERT, RoBERTa) will be integrated into a shared latent space, enabling text generation through forward and reverse diffusion processes. The model will be trained and tested on cleaned versions of the SLAKE and Kvasir-VQA datasets, and its performance will be evaluated using F1-score and BLEU-1 metrics. The goal is to achieve over 85% accuracy on these metrics and to identify the optimal configuration that provides at least a 5% performance improvement among encoder–attention configurations. The explainability of the model will be qualitatively examined through attention heatmaps and CLIP-based similarity analyses.

The management structure will consist of model development, data preprocessing, performance analysis, and reporting phases under the coordination of the project lead. The hardware infrastructure will be supported by GPU-based high-performance computing resources. The results obtained from the project will directly contribute to medical visual question answering, clinical decision support, and multimodal artificial intelligence research. The model's open-source code and evaluation protocols will be shared with the research community, establishing a reusable infrastructure at both national and international levels. The findings will be published in high-impact conferences or journals, and in the long term, the project will pave the way for the development of explainable AI solutions for physician assistant systems, radiology reporting, and clinical education tools. Thus, DiffGenMed-VQA will lay the groundwork for the development of domestic, transparent, and reliable artificial intelligence models in the field of health informatics.

Keywords: Medical artificial intelligence, visual question answering, diffusion-based generative models, multimodal learning, explainable deep modeling

1. RESEARCH CONTRIBUTION

1.1. Significance of the Study, Contribution of the Project and Research Questions

Large Language Models (LLMs) have created a paradigmatic shift in natural language processing (NLP) by representing human language at statistical and semantic levels [1]. These models are categorized into three main classes based on their architectural approaches: (i) autoregressive models that generate text based solely on preceding words (GPT series) [2], (ii) autoencoder models that infer contextual meaning by predicting masked words (BERT derivative models) [3], and (iii) encoder-decoder approaches that offer both text understanding and generation capabilities (T5, BART) [4]. These architectures have provided a universal framework for tasks such as translation, summarization, question answering (QA), and information extraction.

Recently, with the rise of diffusion models, artificial intelligence has expanded into visual and multimodal generation fields, giving birth to a new generative paradigm where random noise is converted into meaningful data structures through probabilistic transformation processes [5]-[7]. This development has strengthened semantic-visual alignment in language-vision representation learning (e.g., Stable Diffusion).

From the perspective of QA systems, the self-attention mechanisms of transformer-based models have made it possible to generate consistent and explainable responses in long contexts [8]. Encoder-based models like BERT and RoBERTa have excelled in extractive QA tasks [3], [9], while generative models like T5 and BART have stood out in open-ended answering. However, EM and F1 metrics, textual similarity statistics (BLEU, ROUGE) [10], and fluency analyses used to evaluate these models can only measure the accuracy dimension of responses; they reflect semantic adequacy, contextual consistency, and explainability dimensions in a limited way.

This deficiency is particularly evident in the medical QA field. The complex structure, terminological density, and high accuracy requirements of medical texts have limited the direct applicability of classical LLMs. Although models like BioBERT [12], ClinicalBERT [13], and PubMedBERT [14] have improved semantic representation in medical texts, they are mostly limited to text-based questions and cannot model visual-text integrity. Large-scale models developed in recent years, such as Med-PaLM [15], have provided doctor-level answering only in the linguistic context, ignoring visual-based clinical contexts. While the field of Medical Visual Question Answering (Med-VQA) is developing in the literature, current approaches are mostly classification-based and limited to predefined answer labels.

The proposed study aims to radically transform existing Med-VQA paradigms with the Diffusion-Based Medical Question Answering Model (DiffGenMed-VQA). Unlike classical classifier-based QA systems, DiffGenMed-VQA builds visual and textual information upon bidirectional conditioning (conditional diffusion). The model (i) encodes medical image and question information in the forward diffusion process with Conditional Gaussian Noising, and (ii) generates the natural language response with the Reverse Diffusion Process. This mechanism allows the model to probabilistically reconstruct both visual semantics and linguistic context.

This project will be one of the first applications of diffusion-based generative modeling in the context of Med-VQA in the literature. The research carries innovation not only at the methodological level but also conceptually and theoretically:

- Combining visual and linguistic representation in a common latent space through diffusion processes is a new information integration method not found in existing transformer-based QA approaches.
- It allows for the development of new criteria such as semantic accuracy, clinical relevance, and visual consistency (e.g., CLIP-based semantic Fidelity) in Medical QA evaluation, rather than just syntactic metrics like EM/F1.
- It endows clinical decision support systems with a near-human interpretation capacity by generating open-ended, explanatory, and evidence-based answers that mimic doctor-assistant interaction.

With these original aspects, DiffGenMed-VQA presents a pioneering framework that establishes synergy between diffusion-based generative modeling and large language models in the field of medical AI.

The fundamental question of this research is formulated as follows:

"Can diffusion-based bidirectional conditioning increase the semantic accuracy, clinical relevance, and explainability of responses in medical visual question answering tasks compared to classical transformer-based approaches?"

As a hypothesis: H_1 : Diffusion-based generative conditioning, unlike traditional encoder-decoder models, generates more consistent, natural, and clinically valid responses by optimizing linguistic context together with semantic features derived from medical images.

This framework will be evaluated both qualitatively and quantitatively. In qualitative analysis, semantic relevance evaluations made by expert physicians will be used, while in quantitative analysis, EM, F1, BLEU, ROUGE, and CLIP-based similarity metrics will be employed. Thus, the study will not only fill the conceptual gap in the literature but also present the first empirical evidence regarding the effectiveness of visual-linguistic diffusion modeling in medical AI systems.

1.2. Research Objectives

The primary aim of the proposed project is to develop a diffusion-based generative model (DiffGenMed-VQA) by transcending the limits of existing classification-based approaches in the field of Medical Visual Question Answering (Med-VQA) [16]. In this context, the study aims to build an explainable, generative, and high-accuracy artificial intelligence system that models the semantic link between medical images and natural language-based questions using bidirectional conditioning. The unique aspect of the project is the design of a new architecture that, unlike classical VQA models, does not limit answers to fixed label sets; instead, it generates justified responses in natural language form through diffusion processes [17]. Thus, the model will strengthen both explainability in clinical information extraction and semantic integrity.

Primary Objectives

1. **To develop a diffusion-based generative Med-VQA model:** A DiffGenMed-VQA architecture capable of producing natural language responses to medical questions by integrating visual (e.g., X-ray, endoscopy) and textual (question descriptions) inputs will be created. The model will represent both language and image information in a shared latent space through conditional forward and reverse diffusion processes [5], [6], [18].
2. **To ensure explainable and justified response generation:** It is aimed that the generated responses are not only accurate but also supported by causal justifications. For this purpose, the model's outputs will be evaluated for explainability by associating them with visual attention heatmaps and textual explanation components [19], [20].
3. **To optimize model performance on qualified datasets:** The goal is for the model, trained on cleaned and balanced SLAKE and Kvasir-VQA datasets, to demonstrate high performance in key metrics [21], [22].
 - F1-Score $\geq 85\%$
 - BLEU-1 $\geq 85\%$ These success levels will provide significant progress compared to current SOTA (State-of-the-Art) Med-VQA models.
4. **To determine the optimal model configuration:** Different design variants will be tested in visual and textual encoder components (e.g., ViT, Swin, CLIP, BERT, RoBERTa) and multi-head attention mechanisms [24]-[27].
 - **Target:** Determining the optimal architectural combination (encoder + attention type + diffusion stage) that provides at least a 5% performance increase in the F1-Score metric.

Expected Results and Feasibility

These objectives are achievable within the duration of the project because:

- The datasets to be used (SLAKE, Kvasir-VQA) are open-access, and their previously modeled and balanced forms are available in the literature [21], [22].
- Existing open-source libraries (Hugging Face Diffusers, OpenAI Glide, Stability AI) and hardware infrastructure (GPU/TPU) are suitable for diffusion-based modeling.
- Performance evaluation is based on clearly defined numerical metrics (F1, BLEU, ROUGE, CLIP-similarity) [10], [28].

In this framework, the proposed project carries the distinction of being one of the first studies to systematically apply generative diffusion modeling in the field of medical visual question answering [16], [18]. Thus, it will present an original contribution to the field of medical artificial intelligence at both conceptual and methodological levels.

2. METHODOLOGY

1. Research on Similar MedVQA projects and Technical Literature

Before actually starting the implementation of the VQA model, to establish the scope of the project a preliminary research on multi model VQA's and medical QA's was needed so literature and similar projects were searched and examined.

1.1 Scope Definition: Establishing scope for MedVQA benchmarks (ImageCLEF MedVQA, VQA-RAD, PathVQA) and multimodal VQA literature; define inclusion criteria (medical domain, vision–language, publicly reported metrics) [55][56][57].

1.2 Literature Review & Extraction: Systematically collecting papers and reports; extracting dataset details (modality, size, licensing), model architectures (encoders/decoders, fusion), and reported metrics from this papers and reports.

1.3 Dataset Inventory: Creating a catalog of candidate datasets with download/usage notes, splits, preprocessing needs, and license constraints; marking for suitability for single vs. multi-modal settings [56][60].

1.4 Metric Assessment: Comparing common metrics for MedVQA (BLEU-n, ROUGE-L, METEOR, CIDEr, BERTScore, accuracy/F1), noting strengths/limitations for clinical QA (e.g., semantic adequacy vs. n-gram overlap) [58][61][62].

1.5 Transformer Survey: Summarizing text encoders (PubMedBERT, BioMedLM, BioBERT, SapBERT, RoBERTa) and vision encoders (ViT, CLIP-ViT) used in MedVQA; recording reported gains and computational costs from this examples[59][63].

2. Creation of Text and Visual Transformer Pipelines

Within this work package, transformer pipelines specific to the Med-VQA task will be designed and implemented to convert textual and visual data into embedding vectors and to integrate them into a single shared vector space [16], [21]. The process consists of defining target formats, model selection, implementation of data loaders, encoder integration, and technical validation stages.

2.1 Definition of Target Formats and Model Selection

In the initial stage, the embedding dimensions and tokenizer preferences expected by the model are defined. For the text processing pipeline, candidate text encoders are evaluated by considering the terminological density of clinical language [14], [17]. Texts are processed using a WordPiece tokenizer compatible with the BERT architecture [3], [17], which reduces out-of-vocabulary (OOV) issues in medical terminology (e.g., *bronchoalveolar*) [29], [30]. On the visual side, Vision Transformer (ViT)–based candidates [1], [4] (such as ViT-B/16, CLIP-ViT, etc.) are evaluated, and the architecture that best represents global context is selected [23], [25].

2.2 Implementation of Data Loaders

Customized data loaders are implemented for the selected models.

- **Text Loaders:** Inputs are tokenized, [CLS] and [SEP] tokens are added, and sequences are truncated or padded according to the maximum sequence length [1]. In addition, attention masks are generated to ensure contextual awareness [17].
- **Visual Loaders:** Images are resized to 224×224 pixels, converted into tensor format, and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$) [34]. Data augmentation operations that preserve semantic integrity (e.g., rotation, contrast adjustments) are integrated into the loader pipeline at this stage.

2.3 Encoder Integration and Interface Wrapping

The selected transformer backbones are wrapped with a consistent interface to ensure structural coherence within the model architecture. This integration standardizes the forward signatures and cross-attention mechanisms of the text and image encoders, enabling textual and visual representations to be seamlessly fused within the DiffGenMed-VQA architecture [16], [18].

2.4 Pipeline Validation and Sanity Suite

To ensure the robustness of the developed pipelines, a comprehensive technical validation set (sanity suite) is established. This process includes:

- **Shape Checks:** Verifying that the generated embeddings and masks fully match the configured dimensions.

- **Embedding Norm and Value Checks:** Ensuring that no NaN/Inf values occur in the vector space and that norm distributions fall within expected ranges.
- **Batch Throughput:** Testing and optimizing the system's data processing speed.

At the end of this work package, technically validated, error-free text–image processing pipelines that produce reliable embeddings and are ready for the fusion stage [40], [41] will be obtained.

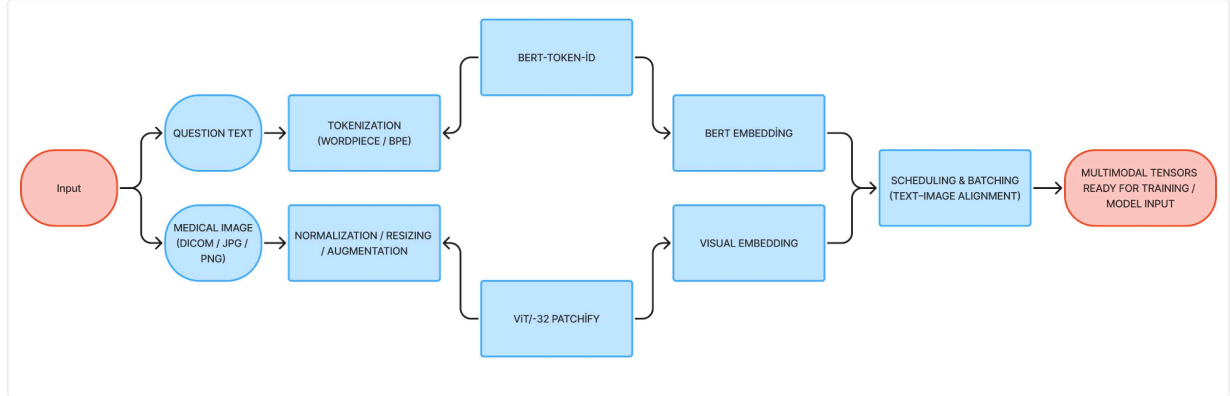


Figure 1. General Stage of Text Processing and Image Processing Pipeline Setup.

3. Diffusion Model Pipeline Integration

This work package aims to establish the training infrastructure of the diffusion model that forms the core of DiffGenMed-VQA, integrate existing data pipelines, and ensure performance optimizations. The process consists of defining training objectives, building the training module, pipeline integration, implementing sampling functions, and activating reliability measures (failsafes).

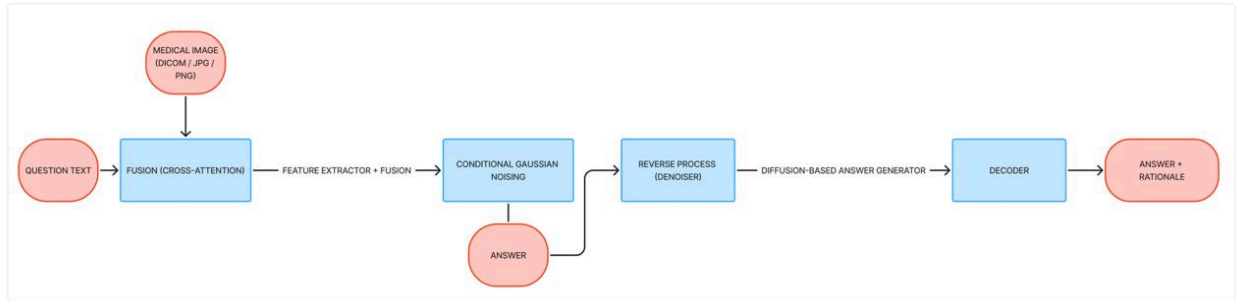


Figure 2. Overall Workflow of the Diffusion Model

3.1 Training Objectives and Configuration Schema

The mathematical foundation of the system is based on the Denoising Diffusion Probabilistic Model (DDPM) framework [1], [3], [4] [44]. While defining the training objectives, a sequential noise process (β_t) is specified over the latent representation of the answer sequence (x_0). The configuration schema includes loss functions, checkpointing strategies, and noise schedules. DiffGenMed-VQA experiments with both linear and cosine-based β_t functions to optimize the noise–diversity trade-off [45], [46].

3.2 Implementation of the Training Module

To translate the theoretical framework into practice, a comprehensive diffusion training loop is constructed. This module manages model initialization, noise schedule handling, loss computation, and optimization steps. To ensure reproducibility of experiments, deterministic seeding mechanisms are integrated into the codebase. In addition, an automatic checkpointing system is established to regularly save the model state during training.

Integration and Fusion of Existing Pipelines

Textual and visual embeddings generated in previous stages are connected to the conditioning mechanism of the diffusion model. This integration covers incorporating existing data loaders, batching systems, and masking structures into the diffusion network. At the core of this integration lies the Conditional Information Gaussian Noising (CIGN) approach. This structure follows a configuration-driven fusion strategy, injecting conditional information into the model in both the forward and reverse processes:

$$x_t = \sqrt{a_t}x_0 + \sqrt{1 - a_t}\epsilon + \gamma_t \cdot f(q, v)$$

Here, $f(q, v)$ denotes the integrated multimodal fusion function [47], [48]. Through this mechanism, the system achieves a structure that preserves both linguistic and visual coherence.

3.3 Implementation of Sampling and Evaluation Functions

To assess the model's performance, sampling and validation functions are implemented.

- **Qualitative Analysis:** Sampling scripts are created to evaluate the visual and textual quality of the model's generated outputs.
- **System Metrics:** Logging and profiling tools are integrated to measure system performance metrics such as latency, memory usage, and throughput.
- **Guided Generation:** Using the Classifier-Free Guidance (CFG) method, conditional and unconditional predictions are combined linearly to improve response quality [3], [53].

3.4 Performance and Reliability Measures

To ensure training stability, performance and reliability measures (failsafes) are incorporated into the system:

- **Optimization:** FP16 (half-precision) training and gradient checkpointing options are applied to improve memory efficiency and increase training speed.
- **Error Handling:** OOM-safe batching strategies and automatic error handling/retry mechanisms are developed to address out-of-memory (OOM) situations.
- **End-to-End Tests:** Fast smoke tests covering both training and inference processes are integrated to verify overall system health.

In conclusion, this work package transforms the theoretical foundations referenced in [1], [3] into a concrete software architecture defined across Tasks 3.1–3.5, ensuring that DiffGenMed-VQA can be trained reliably and with high performance.

4. Model Training and Testing

This work package covers the training and evaluation phases of the developed DiffGenMed-VQA architecture. The aim is to ensure the model learns stably under multimodal conditioning, produces contextually consistent responses, and measures medical question-answering performance in a statistically verifiable manner. The training process is built upon a denoising diffusion-based probabilistic learning framework, where both text and visual inputs are fed into the model as conditional information [5]

Training Process and Optimization Strategy

During the training phase, three fundamental components are present at the model input:

- (1) noisy latent representation (x_t)
- (2) timestep embedding (t)
- (3) conditional multimodal feature vector ($f(q, v)$)

where q represents question embeddings and v represents visual embeddings. The model generates the estimated clean latent ($\hat{x}_0$) or noise ($\hat{\epsilon}$) throughout the reverse diffusion process[5].

Optimization during the training process relies on two main components:

1. Denoising Score Matching Loss (L_{DSM}):

$$L_{DSM} = \mathbb{E}_{t, x_0, \epsilon} [\| \epsilon - \epsilon_{\theta}(x_t, t, f(q, v)) \|^2]$$

This term measures the model's noise prediction capability; it ensures the stability of the diffusion process by minimizing the mean squared error between the predicted noise and the actual noise.

[5][43][45]

2. Cross-Entropy Loss (L_{CE}):

For the generated textual response token sequence $y=(y_1, \dots, y_n)$:

$$L_{CE} = - \sum_i \log P_{\theta}(y_i | y_{<i}, f(q, v))$$

This term maximizes the probability of the correct word and maintains linguistic fluency during generation[1][4].

The total loss is expressed as the weighted combination of these two terms:

$$L_{total} = \lambda_1 L_{DSM} + \lambda_2 L_{CE}$$

Here, hyperparameters λ_1 and λ_2 control the generativity-accuracy balance.

Training is conducted with the AdamW optimization algorithm (learning rate $\approx 1e-4$; weight decay=0.01). Cosine annealing scheduler and warm-up phase support stable learning in early steps[7]. Batch normalization, gradient clipping, and mixed precision (FP16) techniques are employed to prevent gradient explosions often seen in diffusion-based models[46]. The training process continues between 200-400 epochs; the mini-batch size (8-16) is adjusted according to GPU memory capacity.

To prevent overfitting, early stopping, dropout ($p=0.1-0.3$), and data augmentation strategies are applied. Additionally, at the end of each epoch, the generalization capability of the model is monitored by calculating the F1-score and semantic consistency loss on the validation set.

Testing the Model and Performance Evaluation

In the testing phase, the model generates open-ended natural language responses for previously unseen image-question pairs. Generation is carried out using classifier-free guidance scaling ($\omega \in [1.0 - 3.0]$) and stochastic sampling (e.g., top-p, nucleus sampling) techniques. Thus, both consistency and diversity are maintained. Performance evaluation is performed at both syntactic (exact) and semantic levels:

- **Exact Match (EM):** The ratio of the generated answer matching the gold standard answer exactly.
- **F1-Score:** Measures the intersection ratio of correct and predicted words; captures partial accuracies.
- **BLEU-1/2:** Evaluates n-gram level word agreement of the model; measures linguistic fluency.
- **ROUGE-L:** Evaluates semantic overlap in long sequences.
- **Semantic Similarity (Cosine / Sentence-BERT):** Measures closeness in semantic space, covering synonym or paraphrase situations.
- **CLIP-Score:** Evaluates the consistency of the response with the visual context (image-text alignment); measures multimodal suitability specific to Med-VQA.

Additionally, expert physician evaluation (expert-in-the-loop validation) is planned during the testing process. Physicians evaluate model responses in three dimensions: accuracy, clinical relevance, and interpretability. Thus, a holistic verification between automated metrics and human judgment is ensured[19],[20],[54].

Post-Training Analysis and Model Versioning

After training is completed, the model achieving the highest F1-score on the validation set is selected using the best-checkpoint strategy. While training logs and metrics are documented using a TensorBoard-based monitoring system, the **TQDM** library is utilized to visualize the training progress and iteration speed in real-time within the console output. Furthermore, the inference performance of the final model (e.g., time per response, GPU memory consumption) is profiled to test its suitability for clinical-scale applications. Consequently, this work package ensures the DiffGenMed-VQA's generative diffusion process is learned correctly, producing contextually strong and semantically consistent responses, and verifying its performance with both statistical and qualitative (expert) criteria. This structure will form a solid foundation for the work to be carried out in subsequent work packages (model improvement, hyperparameter optimization, XAI module).

5. Result Documentation Module

5.1 Objectives and Scope

This work package establishes a systematic framework for recording, organizing, and analyzing experimental results across the entire model development pipeline. The module captures quantitative metrics from training, sampling, and evaluation phases to enable comprehensive performance comparison across different model configurations.

5.2 Data Collection and Logging Architecture

The documentation module collects metrics across three key phases:

- **Training Phase Metrics:**

Independent variables: learning rate, batch size, diffusion steps, optimizer type, scheduler configuration

Dependent variables: training loss, validation loss, convergence rate, total training time, GPU memory utilization, checkpoint intervals

Hardware context: GPU model, VRAM, compute capability, dataset version

- **Sampling Phase Metrics:**

Independent variables: sampling steps, seed (for reproducibility), temperature/top-p parameters, batch size, model checkpoint used

Dependent variables: total samples generated, sampling duration, samples per second (throughput), memory peak usage

Quality indicators: sample validity checks, token distribution statistics

- **Evaluation Phase Metrics:**

Independent variables: encoder type (PubMedBERT, BioMedLM, BioBERT, SapBERT, ViT variants), attention mechanism (self-attention, cross-attention, gated fusion), hyperparameter configuration

Dependent variables: overall accuracy, question-type accuracy (Yes/No, Open-Ended), BLEU-1/2/3/4, ROUGE-L, METEOR, CIDEr, BERT Score, F1 Score, semantic similarity, inference latency

5.3 Methodology and Tools

Storage and Schema: All metrics are recorded in a standardized Excel-based schema with incremental append functionality, enabling:

- Non-destructive logging (multiple runs append to the same file)
- Traceability (timestamp, model identifier, dataset version recorded for each entry)
- Query-ability (spreadsheet structure supports filtering and comparative analysis)

Automation and Integration: The documentation module is integrated directly into training (train.py), sampling (efficient_sample.py), and evaluation (eval_DiffuVQA.py) scripts via function hooks, ensuring automatic capture without manual intervention.

Comparative Analysis Methods:

- Tabular comparison of configurations across all metrics
- Metric correlation analysis (e.g., attention patterns vs. accuracy gains)
- Performance trending (convergence speed, sample quality over time)
- Hardware utilization profiling (GPU memory, training time per step)

6. Model Improvement and Hyperparameter Tuning

6.1 Research Objectives and Hypothesis

This work package systematically optimizes model performance through controlled variation of encoder architectures, attention mechanism, hyperparameters. The objective is that domain-specific text encoders (PubMedBERT, BioMedLM, BioBERT, SapBERT) combined with cross-attention fusion and tuned hyperparameters will achieve >5% relative improvement in overall accuracy compared to the baseline (0.80 on SLAKE test set).

6.2 Experimental Design and Independent Variables

Encoder Configuration :

- Text encoders: PubMedBERT (768-dim), BioMedLM (1024-dim), BioBERT (768-dim), SapBERT (768-dim), RoBERTa-Large (1024-dim baseline)
- Vision encoders: ViT-B/32, ViT-L/14, CLIP-ViT-B/32, ResNet50-CLIP (2-3 variants)
- Output standardization: Projection layers normalize all encoders to a common dimension (768 or 1024)

Attention Mechanisms :

- Self-attention: Standard transformer self-attention in diffusion model
- Cross-attention: Text-to-image and image-to-text cross-attention blocks (1-2 layers)
- Sparse attention: Long-range sparse patterns for efficiency

Hyperparameter Space):

- Learning rate: {1e-5, 5e-5, 1e-4} (≥3 levels)
- Batch size: {4, 8, 16} (≥3 levels)
- Attention heads: {4, 8, 16} in cross-attention blocks
- Diffusion steps: {500, 1000, 2000, 4000} in noise schedule
- Dropout: {0.1, 0.2, 0.3}
- Warmup steps: {1000, 5000, 10000}

Experimental scope: Factorial design with ≥6 primary configurations (balanced selection from the full space) + 2 secondary refinement runs based on preliminary results

6.3 Dependent Variables and Evaluation Metrics

Primary Metrics:

- Overall accuracy: Exact match on test set (primary success criterion)
- Question-type accuracy: Yes/No, Open-Ended, Numeric subsets
- Convergence speed: Steps to reach 90% of final validation accuracy
- Sample quality: BLEU-1/2/3/4, ROUGE-L, METEOR, CIDEr scores

Secondary Metrics:

- GPU memory peak: Peak VRAM usage during training and inference
- Training throughput: Steps per second (efficiency)
- Inference latency: Seconds per sample on test set
- Stability: Loss curve variance across repeated runs with same seed
- 6.4 Experimental Protocol and Validation Strategy

6.4 Experimental Protocol and Validation Strategy

Reproducibility Framework:

- Fixed random seed (102) across all runs to ensure deterministic behavior
- Held-out test split: SLAKE test set (unseen during all tuning)
- Validation split: SLAKE validation set (used for early stopping, not hyperparameter selection)
- Cross-validation: Optional 5-fold validation on training set for high-uncertainty configurations

Training and Evaluation Procedure:

1. **Baseline establishment:** Confirm baseline accuracy (0.80) from WP-4 using initial RoBERTa-Large + self-attention config
2. **Configuration sweep:** Train ≥6 distinct configurations for 4k steps each (≥4k steps as per WP-4 baseline)

3. **Metric collection:** Log training/validation loss every 500 steps; compute test metrics post-training
4. Checkpoint management: Save best validation checkpoint and final checkpoint for each configuration
5. Statistical analysis: Compare mean $\pm$ std accuracy and BLEU across configurations; identify statistical significance (if $n \geq 3$ replicates per config)

Performance Attribution:

- Isolate contribution of encoder type: Same attention/hyperparams, vary encoder
- Isolate contribution of attention mechanism: Same encoder/hyperparams, vary attention
- Isolate contribution of hyperparameters: Same encoder/attention, vary hyperparams
- Correlation analysis: Accuracy vs. GPU memory, accuracy vs. throughput (trade-off curves)

Deliverables:

1. Modular encoder/attention registry with ≥ 8 supported configurations
2. ≥ 6 trained model checkpoints with documented hyperparameters
3. Comprehensive comparison Excel file with ≥ 5 worksheets (summary, detailed metrics, accuracy comparisons, hyperparameter impact, training curves)
4. Ranked list of best configurations with reproducible CLI commands
5. Documentation of encoder performance, attention mechanism trade-offs, and hyperparameter sensitivity

Success Criteria:

- Primary: At least three encoder/attention configurations achieve ≥ 0.85 overall accuracy ($>5\%$ relative improvement from 0.80 baseline) on the test set
- Secondary: Best configuration is reproducible (fixed seed, saved config, saved checkpoint) and documented in Excel with clear comparison metrics

7. Explainability (XAI) Module

7.1 Objectives and Hypothesis

Objective: Generate and analyze cross-attention visualizations that justify model outputs and relate attention patterns to performance metrics (accuracy, BLEU, F1) gathered in WP-5.

7.2 Experimental Design

Independent Variables

- Encoder/attention configs: Text encoders (PubMedBERT, BioMedLM, BioBERT, SapBERT, RoBERTa), vision encoders (ViT/CLIP variants), attention types (self vs. cross), fusion modes (gated, concat).
- Sampling/decoding settings: Temperature/top-p, number of sampling steps (for generative outputs).
- Layer/head selection: Which layers/heads to visualize (e.g., last 2 layers, top-K heads by max attention mass).

Dependent Variables

- Attribution quality: Attention mass concentration on relevant image regions and key tokens.
- Task metrics: Overall accuracy, question-type accuracy, BLEU-1/2/3/4, ROUGE-L, METEOR, CIDEr, F1.
- Efficiency: Overhead of hook-based capture (latency/memory), visualization generation time.

7.3 Data Collection and Instrumentation

- Attention capture (Task 7.1): Register forward hooks on cross-attention blocks during inference; save tensors plus metadata (sample ID, layer, head, encoder/attention config, seed).
- Provenance: Store run config (encoder, attention type, checkpoint ID), dataset split, and random seed with each capture.
- Storage: Write attention tensors to disk in a compact format (e.g., npz/pt) with sidecar JSON for metadata.

7.4 Visualization Pipeline

- Preprocessing: Normalize attention maps; aggregate across heads (mean/max) and optionally select top-K heads.
- Image overlay: Resize attention maps to image resolution; create heatmaps and blended overlays.
- Text highlighting: Map token indices to text, highlight tokens with highest attention weights.
- Outputs: PNGs for static overlays; lightweight interactive HTML (e.g., Bokeh/Plotly) for layer/head toggling.

7.5 Metric Comparison Dashboard

- Inputs: Metrics from WP-5 runs (accuracy, BLEU, F1, ROUGE, METEOR, CIDEr) plus attention metadata.

- Views: Tables and plots filtered by encoder, attention type, and hyperparameters; side-by-side comparisons of top-3 configs.
- Exports: Excel and static HTML dashboards; summary and per-config sheets.

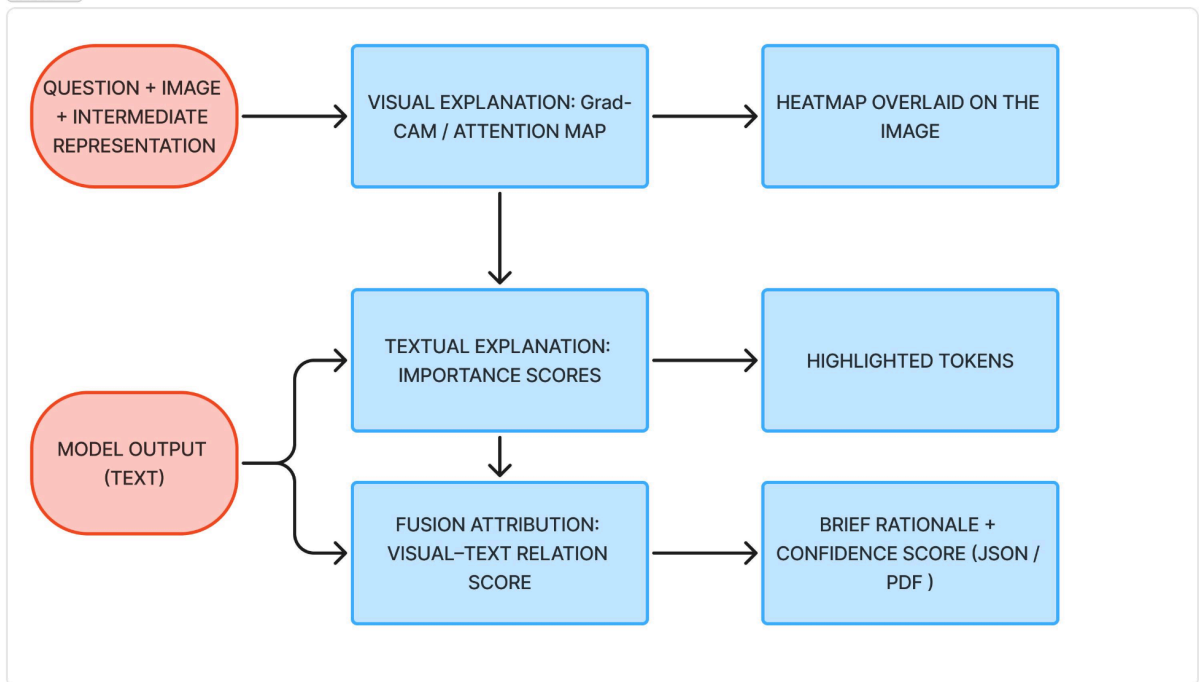
7.6 Interpretability Analysis

- Correct vs. incorrect cases: Contrast attention distributions for correct vs. incorrect predictions; compute attention entropy and focus scores.
- Correlations: Relate attention focus on key regions/tokens to accuracy/BLEU/F1; report simple correlation coefficients.
- Patterns: Document recurring patterns (e.g., higher attention on clinical terms correlates with improved accuracy) and negative cases (e.g., diffuse attention linked to failures).

7.7 Success Criteria

The XAI module generates attention visualizations for ≥ 50 validation samples, produces a comparison dashboard of WP-5 metrics with at least three encoder/attention configurations, and surfaces ≥ 2 actionable insights linking attention patterns to performance differences; all outputs are reproducible with saved configs, seeds, and checkpoints.

Section 1



3 PROJECT MANAGEMENT

3.1 Work Package Timetable

WORK PACKAGE TIMETABLE (*)

WP No	Work Package Name	Start-End Date (...-...Month)	Contributors
1	Research on Similar MedVQA projects and Technical Literature: Expanding Existing Knowledge accumulation about both singular and multi model VQA systems.	1-1	Project Coordinator
2	Text Processing and Image Processing Pipeline Setup: Establishing an infrastructure that merges the outputs of transformer pipelines processing text (BERT/CLIP tokenizer) and images (ViT/CLIP).	1-2	Project Coordinator, Academic Advisor
3	Diffusion Model Pipeline Integration: Establishing a pipeline for training and integrating the diffusion model.	2-4	Project Coordinator, Academic Advisor
4	Model Training and Testing: Training and testing the diffusion model on the SLAKE dataset and establishing a preliminary (baseline) model.	4-5	Project Coordinator
5	Results Documentation Module: Developing a module to document the training and testing outputs of the diffusion model (total loss, accuracy, F1 score).	5-6	Project Coordinator
6	Model Improvement and Hyperparameter Tuning: Improving model accuracy by adjusting transformer settings and increasing performance through the use of different attention mechanisms (such as cross-attention).	6-7	Project Coordinator, Academic Advisor

7	Explainability (XAI) Module: Justifying the model's outputs using XAI techniques (visualization of the model's cross-attention maps) and conducting a comparative analysis of different model metrics (accuracy, BLEU, and F1 scores) obtained in WP-5.	7-8	Project Coordinator
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3.2 Work Packages

WORK PACKAGE #1	
WP Name: Research on Similar MedVQA projects and Technical Literature	
WP Objective: Expanding Existing Knowledge accumulation about both singular and multi model VQA systems.	
WP Tasks 1.1 Studying academic MedVQA studies(ex: ImageCLEF MedVQA, VQA-RAD, PathVQA) 1.2 Searching for usable datasets 1.3 Searching for applicable meaningful metrics (ex: blue_1, f1, bert_score, meteor_score, cider_score)	Responsible People 1.1 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 1.2 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 1.3 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner
Success Criteria: Creating an accumulated knowledge about usable medical datasets, metrics and visual/textual transformers that have been proven to be effective in existing Med-VQA projects.	

WORK PACKAGE #2	
WP Name: Creation of Text and Visual Transformer Pipelines	
WP Objective: Design and implement the initial visual and textual data transforming pipelines which encode medical images and image related questions and answers into embeddings, then fuse them together into a single vector.	
WP Tasks 2.1 defining target formats, tokenizer choices and expected embedding dimensions 2.2 Selecting transformer models: Evaluating candidate text encoders and vision encoders. 2.3 Implementing loaders for images(resize/normalize) and text(tokenize/truncate /mask) 2.4 Integrating Encoders, Wrapping selected transformer backbones with a consistent interface(forward signatures / cross-attention mechanism) 2.5 Validating/Uniting pipelines: Create a small sanity suite (embedding norms, shape checks, batch throughput).	Responsible People 2.1 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 2.2 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 2.3 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 2.4 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 2.5 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner
Success Criteria: Working text and vision transformer pipelines produce valid embeddings (correct shapes, masks, and no NaNs/Infs) and a fused representation of the configured dimension.	

WORK PACKAGE #3
WP Name: Diffusion Model Pipeline Integration
WP Objective: Establishing a pipeline for training and integrating the diffusion model.

WP Tasks 3.1 Defining training objectives, conditional inputs, data interfaces and configuration schema (schedules, losses, checkpoints) 3.2 Implementing Training Module: Build the diffusion training loop (model init, noise schedule, loss, optimizer), checkpointing and deterministic seeding. 3.3 Integrating Existing Pipelines: Connect text/vision embeddings to the diffusion model conditioning, integrating existing dataloaders, batching, masking systems to it and implementing configuration-driven fusion. 3.4 Creation of Sampling/Evaluating Functions: Implement validation metrics, sampling scripts for qualitative outputs, and logging/profiling (latency, memory, throughput) with reproducible runs. 3.5 Implementing Performance & Reliability Failsafes: Add FP16/gradient checkpointing options, OOM-safe batching, error handling/retries, and CI-friendly smoke tests to verify end-to-end training/inference	Responsible People 3.1 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 3.2 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 3.3 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 3.4 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 3.5 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner
Success Criteria: The diffusion pipeline trains end-to-end for 1k steps with the configured text/vision conditioning without errors and successfully generates samples on the validation set.	

WORK PACKAGE #4	
WP Name: Model Training and Testing	
WP Objective: Training and testing the diffusion model primarily on SLAKE dataset and establishing a preliminary (baseline) model.	
WP Tasks 4.1 Preparing Dataset: Arrange and validate SLAKE dataset's train/val/test splits, verify image/text integrity, modify existing data loaders so that they can load SLAKE dataset. 4.2 Setting up Training: Define hyperparameters and config for diffusion training on SLAKE, modify and run the training loop. 4.3 Evaluation & Testing: Modify and run sampling and evaluation scripts using trained models, run end-to-end smoke test to ensure the systems does its intended purpose 4.4 Check Reproducibility & Reliability: Train and evaluate models with both same different hyperparameters using a different dataset than SLAKE.	Responsible People 4.1 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 4.2 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 4.3 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 4.4 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner
Success Criteria: A baseline diffusion model trains end-to-end on SLAKE for at least 4k steps without errors, shows a decreasing validation loss, and successfully generates samples on the validation split with stable throughput.	

WORK PACKAGE #5	
WP Name: Implementing Result Documentation Module	
WP Objective: Developing a module to document the training and testing outputs/ metric result of the model.	
WP Tasks 5.1 Selecting Requirements & Schema: Define a stable metric schema (field names, types, sheet layout, file naming conventions) for each of the steps' (training, sampling, evaluating) 5.2 Implementing Export Module: Create an excess module which creates or writes on existing excel files. 5.3 Integrating Module API & CLI: Expose clear functions/CLI to append evaluation, training, and sampling records; integrate calls inside train.py, eval_DiffuVQA.py, and sampling scripts to auto-log after runs. 5.4 Do Smoke Screen Test: Add unit/smoke tests that create Excel files in a temp folder, verify required columns, numeric formatting, and append behavior; ensure idempotent runs don't corrupt files 5.5 Documentation & Examples: Document usage with copy-paste commands and short code snippets	Responsible People 5.1 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 5.2 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 5.3 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 5.4 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 5.5 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner
Success Criteria: Running baseline train/eval/sampling produces Excel files in the reports directory with the agreed schema, passing automated tests for file creation, column consistency, and append behavior.	

WORK PACKAGE #6	
WP Name: Model Improvement and Hyperparameter Tuning	
WP Objective: Improving model accuracy by adjusting transformer settings, by using different transformers and increasing performance through the use of different attention mechanisms (such as cross-attention).	
WP Tasks 6.1 Trying Different Encoders: Refactor text/vision encoder loading into a registry or factory pattern with standardized output dimensions; support swapping PubMedBERT/BioMedLM/BioBERT/SapBERT and add 2-3 vision transformer options (ViT variants, CLIP-ViT) via configuration. 6.2 Exploring Attention Mechanisms: Implement cross-attention fusion between text and image embeddings; provide config toggles for self-attention, cross-attention, sparse-attention and gated fusion 6.3 Defining and Testing a Hyperparameter Grid: Define a searchable parameter space (learning rate, batch size, attention heads, diffusion steps, dropout, warmup) and run a baseline sweep (≥6 configurations) to identify top-performing settings; log all runs.	Responsible People 6.1 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 6.2 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 6.3 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 6.4 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner

6.4 Comparing: Establish a reproducible evaluation protocol (fixed val/test splits, metrics, seeds) to compare results with each other	
Success Criteria: At least three encoder/attention configurations achieve ≥ 0.85 overall accuracy ($>5\%$ relative gain) on the test set with documented reproducible configs and saved checkpoints; the best configuration is clearly identified with comparison metrics logged in Excel reports.	

WORK PACKAGE #7	
WP Name: Implementing An Explainability (XAI) Module	
WP Objective: Justifying the model's outputs using XAI techniques (visualization of the model's cross-attention maps) and conducting a comparative analysis of different model metrics (accuracy, BLEU, and F1 scores) obtained in WP-5.	
WP Tasks 7.1 Attention Map Extraction: Implement hooks to capture cross-attention weights between text/image embeddings during inference; save attention tensors to disk with metadata for offline analysis. 7.2 Visualization Module: Build a visualization pipeline that overlays attention maps on input images, highlights attended text tokens, and generates heatmaps; output annotated images and interactive HTML reports for qualitative inspection. 7.3 Metric Comparison Dashboard: Create automated comparison tables and plots across all WP-5 configurations; support filtering by encoder type, attention mechanism, and hyperparameters; export to Excel and static HTML. 7.4 Interpretability Analysis: Provide tools to identify high-attention regions for correct vs. incorrect predictions; quantify attention-metric correlations; document patterns.	Responsible People 7.1 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 7.2 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 7.3 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner 7.4 Ahmet Barış Özel, Ali Ekin Özçetin, Mert Güner
Success Criteria: The XAI module successfully generates attention visualizations for ≥ 50 validation samples, produces a comprehensive comparison dashboard of WP-5 metrics with at least three encoder/attention configurations, and identifies at least two actionable insights linking attention patterns to performance differences.	

3.3 Risk Analysis

RISK MANAGEMENT TABLE*

WP No	Risk Definition	Precautions (Plan B)
2	Transformer Architecture Incompatibility: The selected BERT/ViT backbones may fail to capture domain-specific medical nuances or produce misaligned embedding spaces.	Alternatively, DeBERTa-v3 (for the text encoder) and BLIP-2 or LLaVA (for vision-language modeling) architectures will be evaluated. Hyperparameter optimization will be performed before changing the model architecture.

3	Technical issues such as tensor (vector) dimensionality mismatches may arise during the integration of the diffusion model into the pipeline established in WP-2-3.	Implement strict unit tests for tensor shapes at each layer. Utilize Gradient Clipping and manual logging of noise schedules to identify bottlenecks in the diffusion loop.
4	Data Quality & Overfitting on SLAKE: Potential for noisy labels or corrupted images in the SLAKE dataset leading to poor generalization.	Apply automated data cleaning scripts and anomaly detection. Use Cross-Validation and data augmentation (rotation, contrast adjustment) to ensure the model isn't just memorizing the baseline.
5	Reporting Module Failure/Data Corruption: Automated logging/filing may fail during long training runs, leading to lost metrics or corrupted Excel files.	Implement Idempotent writing (check before write) and a backup logging system (like Weights & Biases or local CSV mirrors) to ensure results are captured even if the Excel export fails.
6	Hardware Resource Constraints: GPU Memory (VRAM) limitations during multi-model hyperparameter sweeps or high-resolution ViT training.	Enable Gradient Accumulation , FP16 (Mixed Precision) , and Gradient Checkpointing. If local resources fail, migrate workloads to Google Colab Pro or AWS/Azure instances.
7	XAI Interpretability Gaps: Cross-attention maps may be too "noisy" or diffuse to provide meaningful medical insights, or hooks may significantly slow down inference.	Implement Grad-CAM as an alternative to raw attention maps. Use Layer-wise Relevance Propagation (LRP) to refine heatmaps and optimize hooks to run only on a validation subset.

4. PROJECT OUTCOMES

Output Type	Output	Expected Time of Output
Scientific/Academic (Conference Paper, Journal Paper, Book Chapter etc.)	1. Upon completion of the project, it is planned that the results will be published in the Dokuz Eylül University Journal of Engineering and Natural Sciences and presented as a paper at an International Computer Science and Engineering Conference .	1. 6 - 8 Month
Economical/Commercial/Social (Product, Prototype, Patent etc.)	1. At the end of the project, the developed DiffGenMed-VQA model will provide an open-source and domestically developed medical visual question answering infrastructure, contributing to the localization of AI-based diagnostic decision support systems in the field of health informatics in Türkiye. 2. The open-source version of the model will be usable by university hospitals and R&D units; in the long term, this will establish a high economic value foundation for medical reporting, clinical education, and decision support systems.	1. 6 - 8 Month 2. 6 - 8 Month

5. ALIGNMENT WITH UN SUSTAINABLE DEVELOPMENT GOALS

Impact Area	Does project impact? Direct or indirect?	Impact Definition
Social How the project affects people and society (Who benefits from the project? Does it improve accessibility, inclusion, or equality? Does it reduce manual effort or improve quality of life?) SDG 10 - Reduced Inequalities	Indirect	The project aims to make the model's open-source code and evaluation protocols available to the research community, creating a reusable national and international infrastructure. By developing a "physician assistant system" and "clinical education tool", it improves accessibility to high-level medical interpretation capabilities, potentially aiding junior doctors or clinics with fewer resources.
Health How the project affects health directly or indirectly (Does it reduce stress, workload, or human error? Does it support health monitoring, safety, or wellbeing? Does it prevent unsafe working conditions?) SDG 3 - Good Health and Well-being	Direct	The primary goal is to develop a system for "clinical decision support" that mimics doctor-assistant interaction to provide evidence-based answers. By aiming for high accuracy (F1-Score $\geq 85\%$), and reducing semantic uncertainty, the project directly supports accurate diagnosis and reduces the workload and potential for human error in medical analysis.
Security How security is achieved in the project, or how the project increases the security (How is user data protected? What security risks exist?) SDG 16 - Peace, Justice, and Strong Institutions	Indirect	The project heavily focuses on "Explainable AI (XAI)" to ensure the system is "transparent, reliable, and clinically interpretable". It uses "expert-in-the-loop validation" to ensure outputs are trustworthy and clinically relevant, thereby mitigating the risk of "black-box" AI errors in sensitive medical contexts.
Economic How the project contributes to the economy (Does it reduce costs? Does it improve productivity?) SDG 8 - Decent Work and Economic Growth	Direct & Indirect	The project aims to create a "domestic, transparent, and reliable" AI model, contributing to the localization of health informatics technologies which has "high economic value". It enhances productivity by automating the initial analysis of medical images, serving as an assistant to reduce the manual workload of healthcare professionals.

Sustainability and Environment How the project contributes to the sustainability and affects the environment (Does it reduce paper usage? Does it optimize energy or resource consumption? Is computer system usage efficient?) SDG 11 - Sustainable cities and communities SDG 13 - Climate Action SDG 12 - Responsible Consumption and Production	Indirect	The project implements specific technical optimizations to ensure efficient resource usage. Techniques such as "Mixed Precision (FP16)" training, "Gradient Clipping," and "Early Stopping" are used to prevent resource waste and optimize GPU usage. This contributes to responsible consumption of computational energy.
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REFERENCES

EK-1: KAYNAKLAR

- [1] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL-HLT.
- [2] Radford, A., Wu, J., Child, R., et al. (2019). Language Models are Unsupervised Multitask Learners. OpenAI.
- [3] Liu, Y., Ott, M., Goyal, N., et al. (2019). RoBERTa: A Robustly Optimized BERT Pretraining Approach. arXiv:1907.11692.
- [4] Raffel, C., Shazeer, N., Roberts, A., et al. (2020). Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer (T5). JMLR, 21(140).
- [5] Ho, J., Jain, A., & Abbeel, P. (2020). Denoising Diffusion Probabilistic Models. NeurIPS.
- [6] Nichol, A., & Dhariwal, P. (2021). Improved Denoising Diffusion Probabilistic Models. ICML.
- [7] Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-Resolution Image Synthesis with Latent Diffusion Models (Stable Diffusion). CVPR.
- [8] Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention Is All You Need. NeurIPS.
- [9] Yang, Z., Dai, Z., Yang, Y., et al. (2019). XLNet: Generalized Autoregressive Pretraining for Language Understanding. NeurIPS.
- [10] Lin, C. Y. (2004). ROUGE: A Package for Automatic Evaluation of Summaries. ACL Workshop.
- [12] Lee, J., Yoon, W., Kim, S., et al. (2020). BioBERT: A Pre-trained Biomedical Language Representation Model for Biomedical Text Mining. Bioinformatics.
- [13] Alsentzer, E., Murphy, J. R., Boag, W., et al. (2019). Publicly Available Clinical BERT Embeddings. arXiv:1904.03213.
- [14] Gu, Y., Tinn, R., Cheng, H., et al. (2021). PubMedBERT: Domain-Specific Language Model for Biomedical Text Mining. NAACL.
- [15] Singhal, K., Tu, T., Gottweis, J., et al. (2023). Large Language Models Encode Clinical Knowledge. Nature.
- [16] Liu, Bing, et al. "Redefining medical visual question answering using conditional generative diffusion models." Biomedical Signal Processing and Control 111 (2026): 108222.
- [17] Kabir, Raihan, Naznin Haque, and Md Saiful Islam. "A comprehensive survey on visual question answering datasets and algorithms." arXiv preprint arXiv:2411.11150 (2024).
- [18] Ma, Haokai, et al. "Multimodal conditioned diffusion model for recommendation." Companion Proceedings of the ACM Web Conference 2024. 2024.
- [19] Selvaraju, R. R., Cogswell, M., Das, A., et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. ICCV.
- [20] Doshi-Velez, F., & Kim, B. (2017). Towards A Rigorous Science of Interpretable Machine Learning. arXiv:1702.08608.
- [21] Liu, F., Wu, H., Zhao, Y., ure S et al. (2021). SLAKE: A Semantically-labeled Knowledge-enhanced Dataset for Medical Visual Question Answering. arXiv:2102.09542.
- [22] Gautam, Sushant, Michael Riegler, and Pål Halvorsen. "Kvasir-VQA-x1: A Multimodal Dataset for Medical Reasoning and Robust MedVQA in Gastrointestinal Endoscopy." MICCAI Workshop on Data Engineering in Medical Imaging. Cham: Springer Natwitzerland, 2025.
- [23] Dong, Wenjie, et al. "Generative models in medical visual question answering: A survey." Applied Sciences 15.6 (2025): 2983.
- [24] Dosovitskiy, Alexey. "An image is worth 16x16 words: Transformers for image recognition at scale." arXiv preprint arXiv:2010.11929 (2020).
- [25] Liu, Z., Lin, Y., Cao, Y., et al. (2021). Swin Transformer: Hierarchical Vision Transformer using Shifted Windows. ICCV. [26] Radford, A., Kim, J. W., Hallacy, C., et al. (2021). Learning Transferable Visual Models From Natural Language Supervision (CLIP). ICML.
- [27] Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention Is All You Need. NeurIPS.
- [28] Hessel, J., Schofield, A., & Lee, L. (2021). CLIPScore: A Reference-free Evaluation Metric for Image Captioning. EMNLP.
- [29] Schuster, M., & Nakajima, K. (2012). Japanese and Korean Voice Search. IEEE ICASSP.
- [30] Wu, Y., Schuster, M., et al. (2016). Google's Neural Machine Translation System: Bridging the Gap Between Human and Machine Translation. arXiv:1609.08144.
- [31] Lee, J., Yoon, W., Kim, S., et al. (2020). BioBERT: A Pre-trained Biomedical Language Representation Model for Biomedical Text Mining. Bioinformatics.
- [32] Larson, David B., et al. "Performance of a deep-learning neural network model in assessing skeletal maturity on pediatric hand radiographs." Radiology 287.1 (2018): 313-322.
- [33] Lundervold, A. S., and A. Lundervold. "An overview of deep learning in medical imaging." Frontiers

in Neuroscience 12 (2019): 65.

- [34] Deng, J., et al. (2009). ImageNet: A Large-Scale Hierarchical Image Database. CVPR.
- [35] Shorten, C., & Khoshgoftaar, T. M. (2019). A Survey on Image Data Augmentation for Deep Learning. Journal of Big Data.
- [36] Perez, L., & Wang, J. (2017). The Effectiveness of Data Augmentation in Image Classification Using Deep Learning. arXiv:1712.04621.
- [37] Khan, S., Naseer, M., et al. (2022). Transformers in Vision: A Survey. ACM Computing Surveys.
- [38] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition (ResNet). CVPR.
- [39] Tan, M., & Le, Q. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. ICML.
- [40] Tsai, Y. H. H., Bai, S., & Morency, L. P. (2019). Multimodal Transformer for Unaligned Multimodal Language Sequences. ACL.
- [41] Liang, Y., & Hu, X. (2022). Late Fusion Approaches for Multimodal Medical AI. Medical Image Computing and Computer Assisted Intervention (MICCAI).
- [42] Chen, Xiaolan, et al. "From visual question answering to intelligent AI agents in ophthalmology." British Journal of Ophthalmology (2025).
- [43] Ho, J., Jain, A., & Abbeel, P. (2020). Denoising Diffusion Probabilistic Models. NeurIPS. [44] Song, Y., & Ermon, S. (2021). Score-Based Generative Modeling Through Stochastic Differential Equations. ICLR. [45] Nichol, A., & Dhariwal, P. (2021). Improved Denoising Diffusion Probabilistic Models. ICML.
- [46] Karras, T., Aittala, M., Laine, S., et al. (2022). Elucidating the Design Space of Diffusion-Based Generative Models. NeurIPS.
- [47] Ramesh, A., Pavlov, M., Goh, G., et al. (2022). Hierarchical Text-Conditional Image Generation with CLIP Latents. arXiv:2204.06125.
- [48] Saharia, C., Chan, W., Ho, J., et al. (2022). Imagen: Photorealistic Text-to-Image Diffusion Models. ICML.
- [49] Peebles, W., & Xie, S. (2023). Scalable Diffusion Models with Transformers. CVPR.
- [50] Kingma, D. P., & Welling, M. (2014). Auto-Encoding Variational Bayes. ICLR.
- [51] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. MICCAI.
- [52] Al Ghallabi, Wafa, et al. "MPromer: A Unified Diffusion-Based Framework for Scalable and Generalizable Multi-Modal Medical Image Segmentation." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2025.
- [53] Ho, J., & Salimans, T. (2022). Classifier-Free Guidance for Diffusion Models. arXiv:2207.12598.
- [54] Liu, Yushu, et al. "Improving Explainability and Integrability of Medical AI to Promote Health Care Professional Acceptance and Use: Mixed Systematic Review." Journal of Medical Internet Research 27 (2025): e73374.
- [55] ImageCLEF MedVQA overview papers (2019–2022).
- [56] Lau et al., "VQA-RAD: Visual Question Answering in Radiology," 2018.
- [57] He et al., "PathVQA: Pathology VQA dataset," 2020.
- [58] Banerjee & Lavie, "METEOR: An automatic metric for MT," 2005.
- [59] Beltagy et al., "BioBERT," 2019; Alsentzer et al., "ClinicalBERT," 2019.
- [60] Dataset documentation for ImageCLEF MedVQA and VQA-RAD challenges.
- [61] Papineni et al., "BLEU," 2002.
- [62] Zhang et al., "BERTScore," 2020.
- [63] Dosovitskiy et al., "ViT," 2020; Radford et al., "CLIP," 2021.